

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0123

Darrah 138 kV
April 17, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0123. The Interconnection Customer proposes a 116.0 MW natural gas combined cycle facility interconnecting at the Darrah 138 kV substation in the West Virginia area. The request was received on April 08, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 104 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$73,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Gavin Power Llc to Sporn 230 KV L
Contingency: Flatlick to International Paper 500 KV
Loading: 100.1% (Rating: 1,039 MVA)
DFAX: 5.3%
Pre-project loading: 99.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gavin Power Llc to Sporn 230 KV L	Reconductor line	\$6,351,000	1% (\$73,000)
	Total allocated to project		\$73,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.